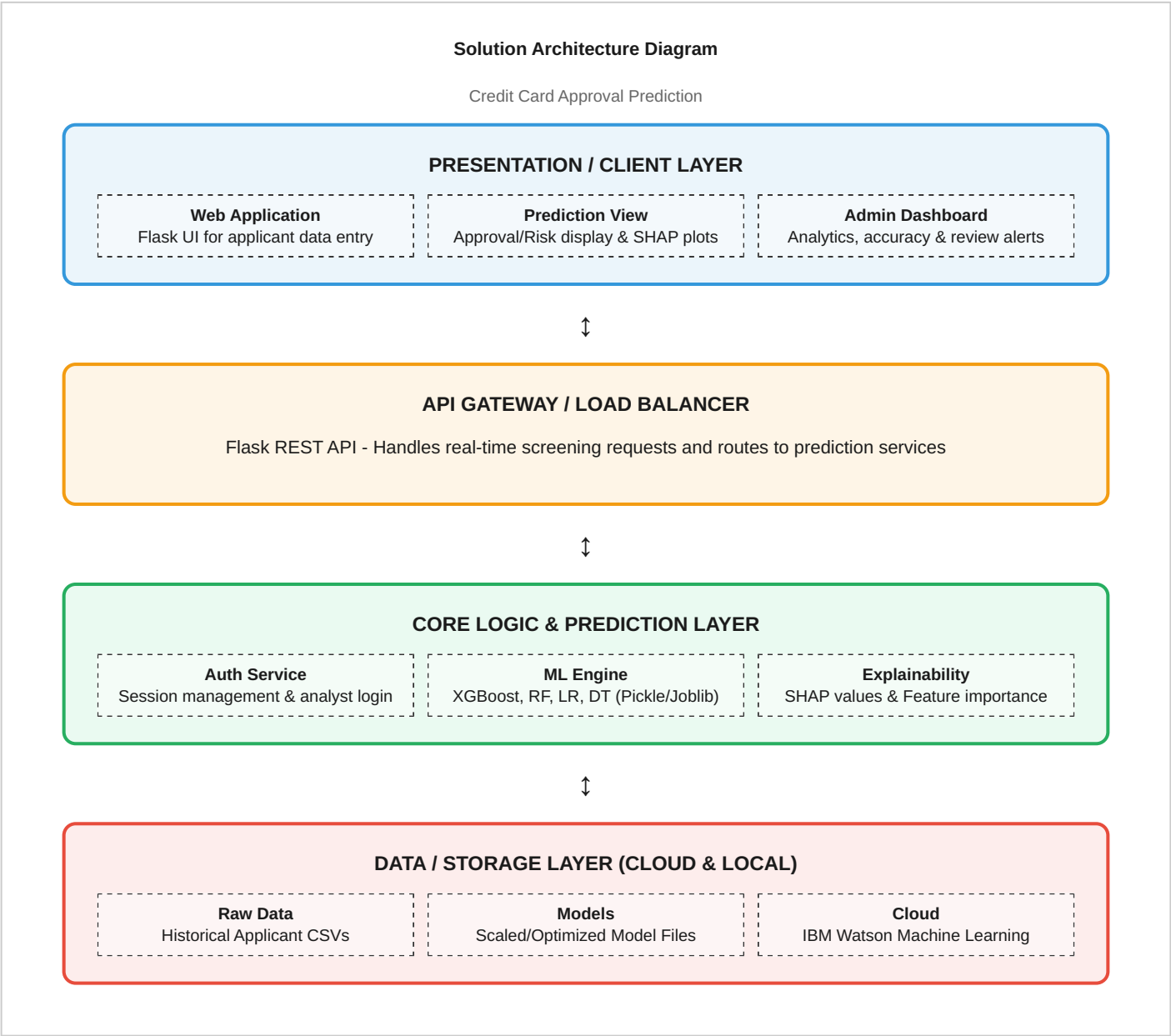


Solution Architecture

Date	14 July 2026
Team ID	
Project Name	Automated Credit Card Approval Screening using ML
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role	Technologies Used
Presentation Layer	Collects financial profiles (income, employment, history) from customers and analysts; displays real-time results and analytics dashboards.	HTML5, CSS3, Bootstrap 5, Flask (Jinja2), Matplotlib
API Gateway	Acts as the intermediary for data transmission between the UI and the ML models; manages endpoint routing and security.	Flask REST API, Flask-JWT (for analyst auth)
Core Logic Service	Preprocesses input data (scaling/normalization), executes one of the four tuned ML models, and generates SHAP explainability insights.	Python, Scikit-learn, XGBoost, SHAP, NumPy, Pandas
External ML Service	Hosts the finalized best-performing model as a scalable cloud endpoint for on-demand predictions.	IBM Watson Machine Learning
Database / Storage	Stores historical application data for re-training and persists the serialized hyperparameter-tuned model files.	CSV, Joblib, Pickle, Cloud Object Storage